

Shruti A. Panicker

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Seeking full-time opportunities from May 2026 | Open to remote

SUMMARY

Data scientist and applied ML engineer with 3.5+ years of enterprise analytics experience, specializing in statistical modeling, NLP, and end-to-end ML delivery across the full project lifecycle from data acquisition through model deployment and stakeholder communication.

EDUCATION

M.S. in Artificial Intelligence – ML Concentration, *Northeastern University* *Boston, MA, USA* | Sep 2024 – May 2026
GPA: 3.875/4 | **Coursework:** Foundations of AI, Data Processing for AI, Machine Learning, NLP, Human Computer Interaction

B.Tech in Computer Science and Engineering, *Amrita Vishwa Vidyapeetham* *Coimbatore, India* | Jul 2016 – Aug 2020
GPA: 8.06/10 | **Coursework:** Data Structures, Computational Intelligence, Deep Learning, Data Science

EXPERIENCE

Khoury College of Computer Sciences, Northeastern University

Graduate Teaching Assistant – DS2500: Intermediate Programming with Data *Boston, MA, USA* | Sept 2024 – Apr 2026

- Lead TA for DS2500 across all 4 semesters of MS (250+ students/semester); promoted to curriculum design role, redesigning assignments and project structure adopted department wide.

Mu Sigma Business Solutions

Apprentice Leader *Bangalore, India* | Jan 2024 – May 2024

- Led a **cross-functional team of 10 decision scientists** delivering product analytics for a Fortune 20 technology company's collaboration device portfolio, leading data-driven projects from definition to execution.
- Built and presented a statistical upsell propensity model using Python and SQL-based analysis on large-scale datasets, identifying high-conversion targets through TAM/SAM/SOM segmentation, **driving 5% product adoption growth** through focused campaigns.
- Defined and monitored **25+ product KPIs through an automated dashboard suite** built in Python and SQL, achieving 100% elimination of manual data pulls and enabling data-driven decisions for product leadership.
- Owned full arc from problem scoping to C-suite presentation, partnering with product and business leadership to translate analytical insights into product strategy and investment decisions.

Decision Scientist *Bangalore, India* | Sep 2020 – Dec 2023

- Managed end-to-end ML delivery for a Fortune 20 enterprise portfolio, integrating multimodal data sources to develop and validate predictive models, quantify business impact through KPIs, and deliver reusable analytical frameworks adopted across the enterprise.
- Built **graph-based entity resolution system** using Python to deduplicate Device IDs across large-scale datasets, applying network association analysis to identify anomalous identity patterns and restore data integrity across all downstream portfolio metrics.
- Designed a three-state customer lifecycle framework (inactive/churned/reactivated) using classification and behavioral segmentation across large-scale datasets, enabling targeted re-engagement campaigns and **driving 18% engagement increase in 6 months**.
- Identified product friction points through hypothesis-driven analysis in Python and SQL, joining behavioral usage signals with support data to diagnose drop-off across onboarding stages and recommend targeted interventions, **driving 30% increase in reactivation**.
- Led 8-month EUDB-compliant pipeline migration across 25+ dashboards, redesigning ETL transformation and aggregation layers to ensure EU data residency compliance with no loss in metric accuracy.

PROJECTS

- Semantic Turn Detection for Voice AI** *Northeastern University*
Built an intent-conditioned voice system that classifies utterances and adapts model behavior dynamically, fine-tuning LLM ensemble weights based on whether an utterance is transactional, informational, or reflective. Developed annotation guidelines and human evaluation methodology **achieving Naturalness 6/7, Frustration 1.53/7, Trust 5.7/7 on a 7-point Likert scale**.
- Empathy Detection – WASSA 2024 Shared Task** *Northeastern University*
Fine-tuned BERT for regression across 11K+ conversational data points, predicting empathy, emotional polarity, and intensity. Designed intrinsic and extrinsic evaluation frameworks to validate model behavior, achieving 0.597 average Pearson correlation and **top 4 on the WASSA 2024 shared task leaderboard**.

SKILLS

ML & AI: PyTorch, Keras, TensorFlow, scikit-learn, BERT/Transformers, Hugging Face, NLP, LLM Evaluation, LLM Fine-tuning, Agentic AI, Prompt Engineering, Statistical Modeling, Predictive Modeling, Classification, Regression, Feature Engineering, A/B Testing, Causal Inference, Experimentation, User Behavior Analysis, Cohort Analysis, Funnel Analysis, Responsible AI | **Data Engineering:** Azure Data Factory, ADLS Gen1 & Gen2, Azure Pipelines, Apache Spark, Hadoop, EUDB & GDPR Compliance | **Languages & Query:** Python, SQL, SCOPE, Power BI DAX, C#, R | **Visualization & BI:** Power BI, Tableau, Azure Data Explorer | **Certifications:** AWS Certified Cloud Practitioner (Credential ID – Y9C4WRJCBJVEQ1KL)